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052411-1 Business Intelligence 1 (2026S)

Assignment 1

(Please note: submission deadline: 15.05.2026)

Group 12

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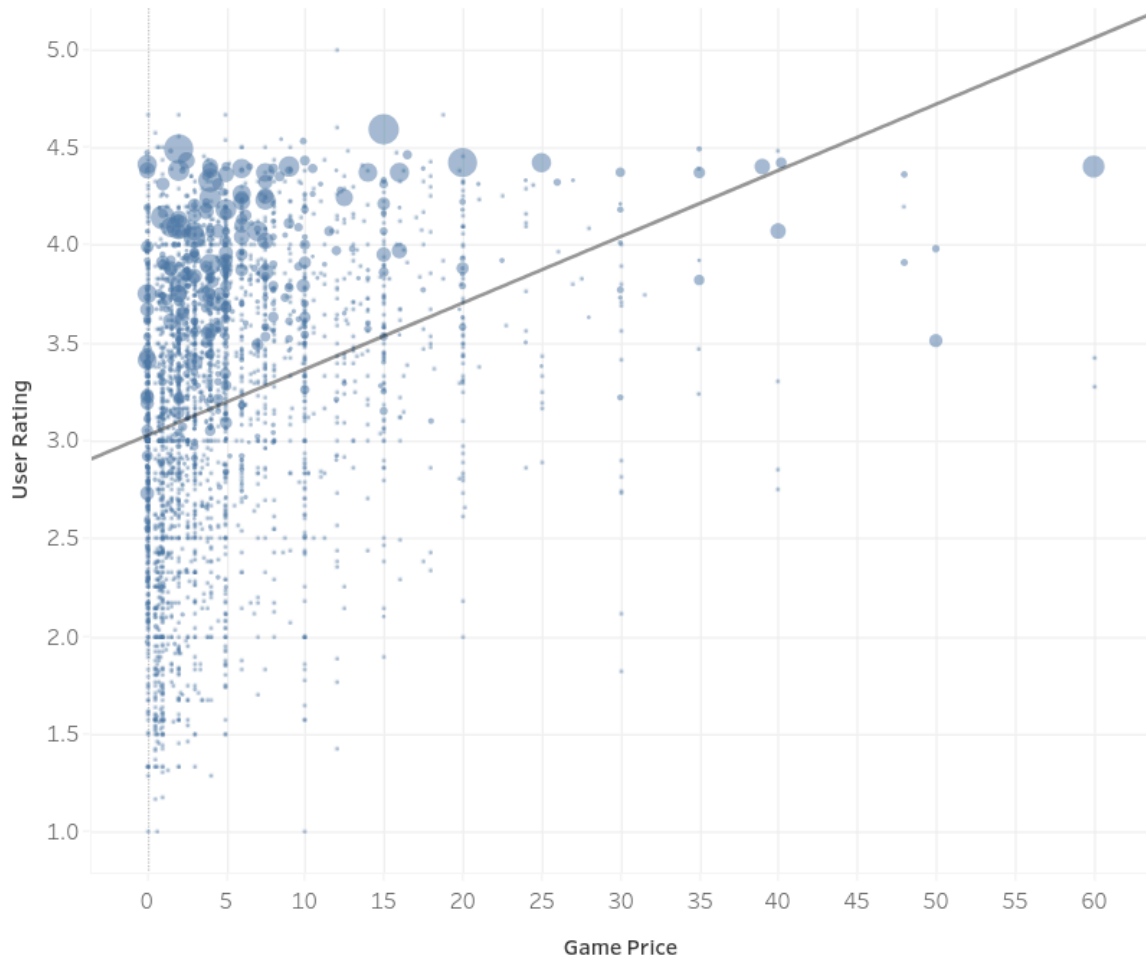
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1. Relationship Between Game Price and User Rating



This plot analyzes the relationship between game price and user rating to understand whether more expensive games tend to receive better ratings from players. Each point in the scatter plot represents a game, using the variables Price and Rating. The trend line helps visualize the overall direction of the relationship between these two variables. From the visualization, most games are concentrated in the lower price range, particularly between \$0 and \$10. This indicates that the majority of games in the dataset are relatively affordable. At the same time, most user ratings fall between 3.0 and 4.5, suggesting that many games are generally well received by players regardless of their price. The trend line shows a slight positive relationship between price and rating. This means that, on average, higher-priced games tend to receive somewhat better ratings. However, the relationship is not very strong because highly rated games can still be found in the low-price range. Several inexpensive games achieve ratings above 4.0, showing that quality is not limited only to premium-priced titles. The plot also reveals

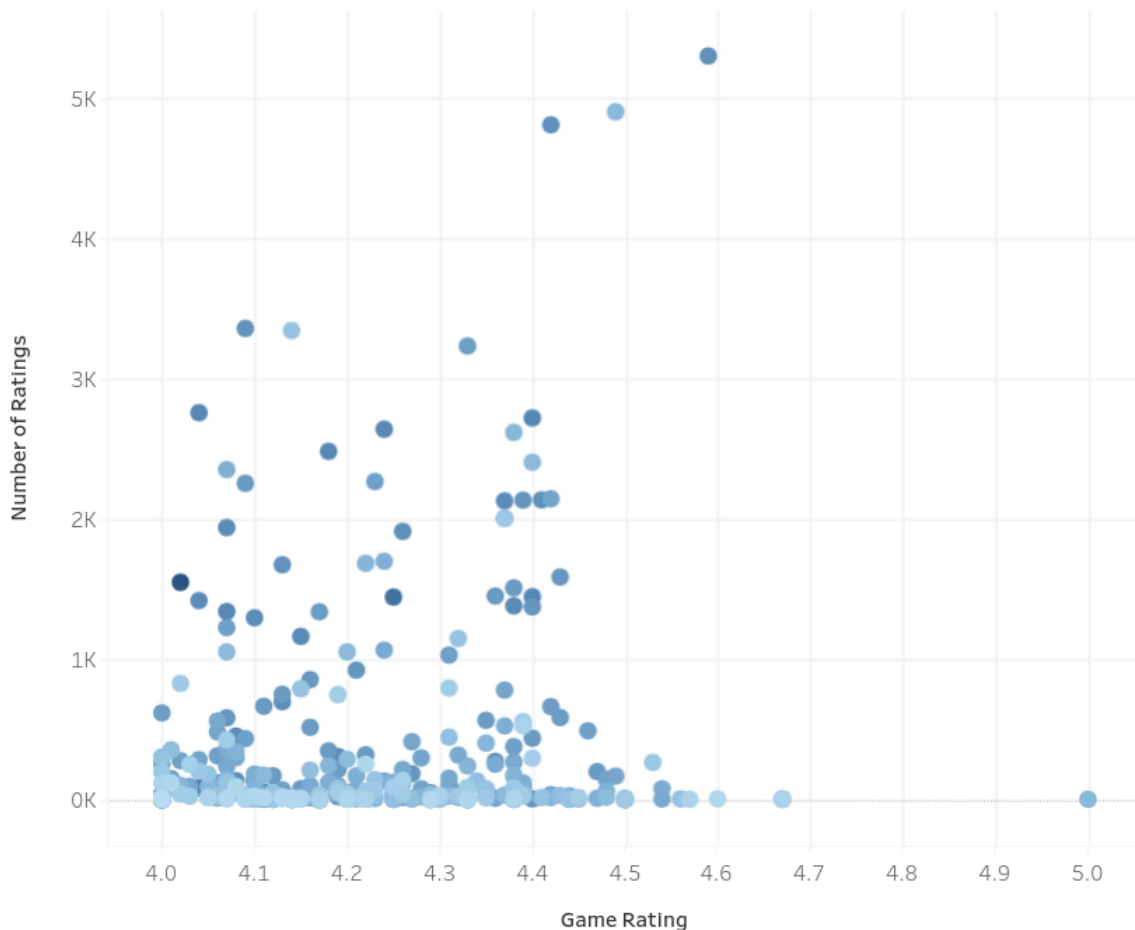
that there are relatively few games in the higher price ranges above \$30–\$60. Because of this smaller number of observations, the relationship between price and rating becomes less consistent for expensive games. Some costly games receive strong ratings, while others remain average.

Overall, the visualization suggests that price alone does not determine user satisfaction. While expensive games may slightly trend toward higher ratings, many affordable games also perform very well. This indicates that factors such as gameplay quality, creativity, graphics, updates, and player experience likely have a stronger influence on ratings than price itself.

Key Insight

Most games are priced below \$10 and still achieve ratings between 3.0 and 4.5. Although the trend line shows a slight positive relationship between price and rating, the correlation is weak. High ratings are found across both low and high price ranges, suggesting that game quality and user satisfaction are not determined solely by price.

2. Hidden Gems: Rating vs Popularity



This plot analyzes the relationship between rating, number of ratings, and suggestion count. The x-axis shows the rating, the y-axis shows the rating count, and the color shows the suggestion count.

From the plot, there is no clear connection between a high rating and a high suggestion count. For example, the only game with a rating of 5.0 has only 7 ratings and around 300 suggestions. So even the highest-rated game is not the most suggested one.

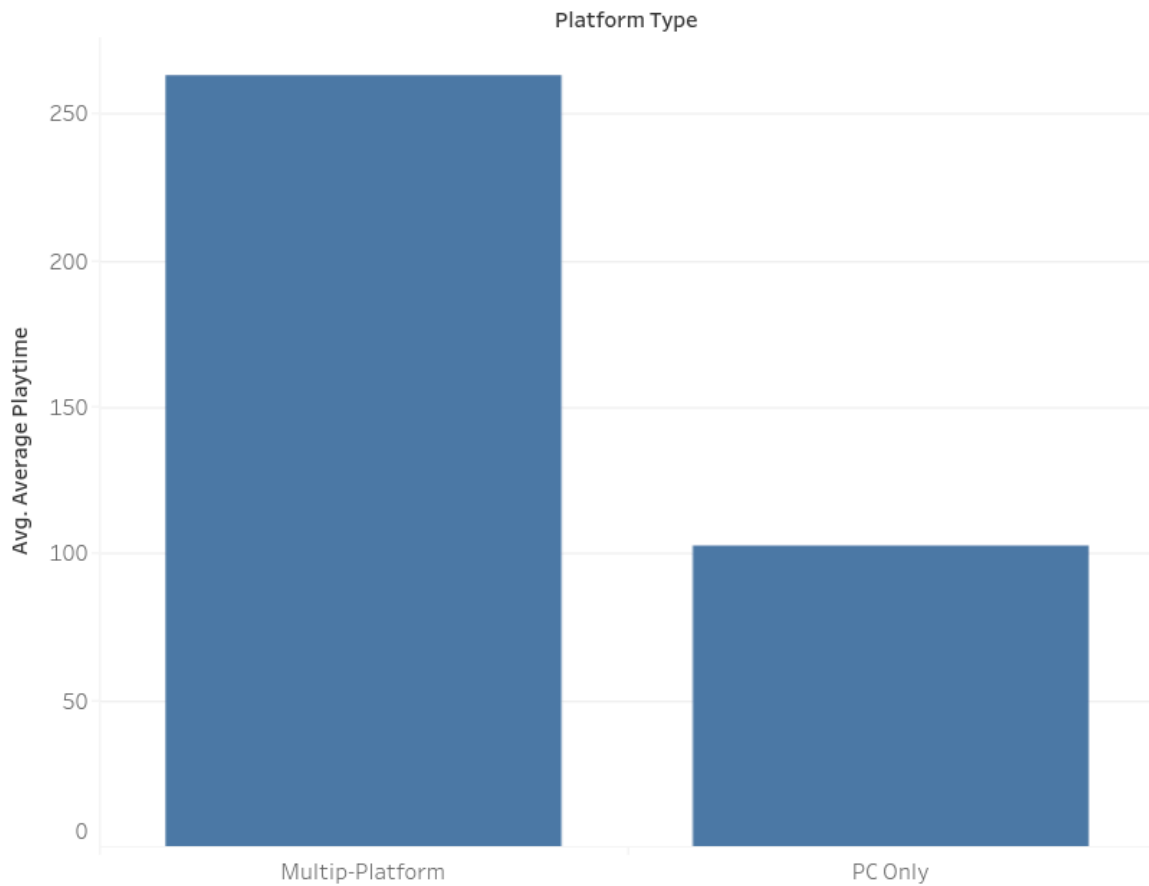
The most rated game has around 5,292 ratings, a rating of about 4.5, and around 624 suggestions. The most suggested game has around 1,135 suggestions, but its rating is only around 4.0. This also shows that the most suggested game is not the highest-rated game.

Overall, higher-rated games are not always suggested more than lower-rated games. Rating count and suggestion count also do not show a very strong connection in this plot. This means that rating, popularity, and suggestions measure different things.

Key insight

Highly rated games are not always suggested more often. The highest-rated game is not the most suggested game, and the most suggested game does not have the highest rating. Therefore, some good games can still remain less visible or act as hidden gems.

3. Average Playtime by Platform Type



This bar chart compares the average playtime between PC-only games and multi-platform games in order to understand how platform availability may influence player engagement. The visualization uses the average playtime values for each platform category to measure how long players continue interacting with games.

The analysis shows a clear difference between the two platform types. Multi-platform games have an average playtime of approximately 262.8, while PC-only games average around 102.7. This indicates that players tend to spend significantly more time playing games that are available on multiple platforms.

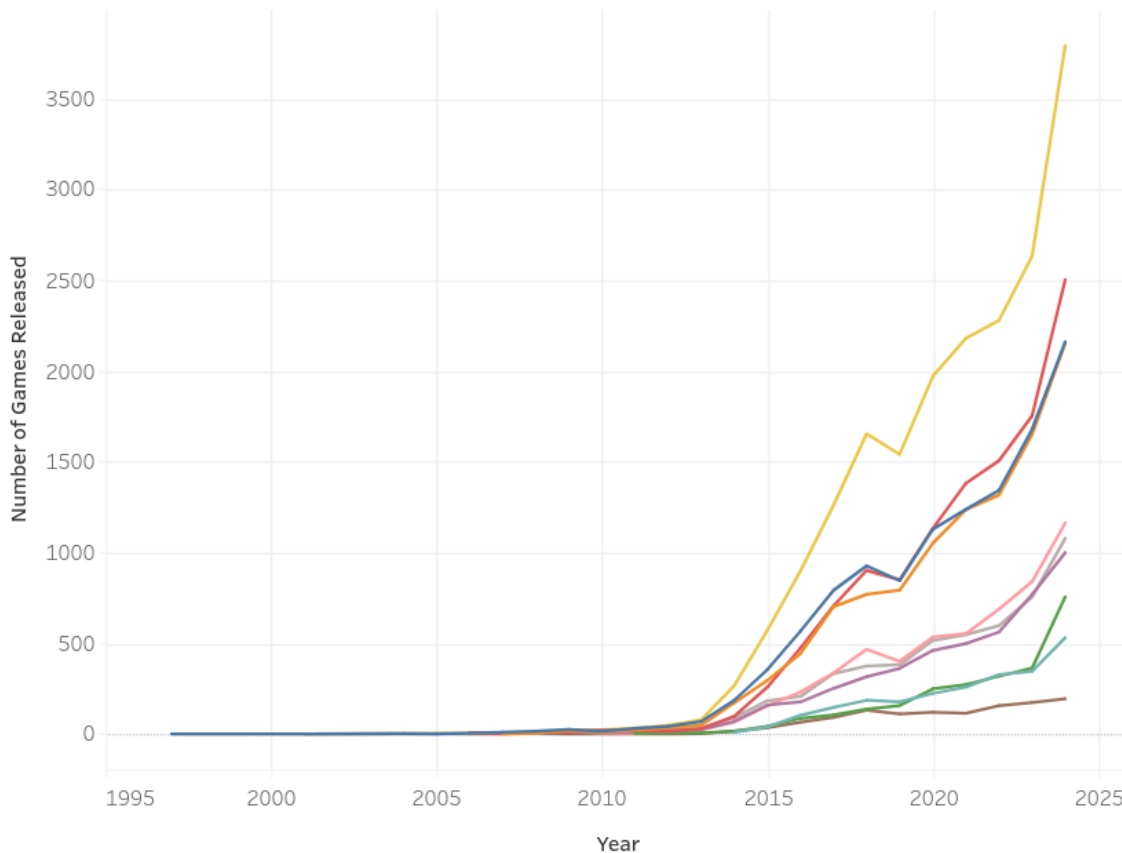
One possible explanation for this difference is that multi-platform games are accessible to a broader audience. Since these games can be played across different devices or systems, they may attract more players and maintain engagement for longer periods. Multi-platform releases may also receive stronger community support, more frequent updates, and better long-term visibility, all of which can contribute to increased playtime.

In contrast, PC-only games show a noticeably lower average playtime. While many PC-exclusive games can still be successful, limiting availability to a single platform may reduce the size of the active player base and overall engagement duration. Overall, the visualization suggests that platform diversity is associated with higher player engagement. Games available across multiple platforms appear to retain players for longer periods compared to games released only on PC.

Key Insight

Multi-platform games achieve an average playtime of approximately 262.8, which is more than double the average playtime of PC-only games (102.7). This suggests that wider platform availability may positively influence player engagement and long-term retention.

4. Growth of Game Genres Over Time



This plot analyzes how the number of released games changed over time for different genres. We look at the years between 1997 and 2024 and count how many games were released per genre in each year.

The clearest result is that Indie games grew the most. Indie was already one of the largest genres in earlier years, but it increased strongly over time and reached its highest value in 2024 with 3,794 released games. So, for our question, Indie is the genre with the strongest growth.

Action, Adventure, and Casual games also increased a lot, but not as strongly as Indie. Their lines move in a similar way, which may be because many games can belong to more than one genre at the same time. A similar pattern can also be seen for Simulation, Strategy, and RPG games.

Free to Play and Early Access also grow over time, but they stay below the larger genres. Sports games stay much more stable and low compared to the others. For example, Sports had around 132 releases in 2018 and 194 releases in 2024, so the increase is small compared to Indie. One possible reason is that sports games are often based on established game series, with new versions released regularly, instead of many completely new games being produced.

Key insight

Indie is the genre that grew the most over time, reaching the highest number of releases in 2024. Action, Adventure, and Casual also show strong growth, while Sports remains low and relatively stable. Overall, the dataset shows that broad genres grew much faster than more specific genres like Sports.